Multimedia Signal Processing - 2nd Module

February, 23 2016

1. **Question (6 pts)**

Consider two discrete memoryless sources X and Y. You observed the output of X and Y obtaining the following sequences:

$$X_1 = \{-1.7,\ -1.6,\ -1.9,\ \ 0.2,\ 0.5,\ 0.3,\ \ 2.2,\ 2.5,\ 2.1,\ -1.9,\ -1.6,\ -1.8,\ 2.2,\ 2.4,\ 2.4,\ 0.5,\ 0.5,\ 0.2,\ \ 2.3,\ \ 0.5\},$$
$$Y_1 = \{-1.5,\ \ 0.4,\ \ 2.4,\ -1.7,\ 0.0,\ 2.3,\ -1.8,\ 0.5,\ 2.3,\ -1.5,\ -1.5,\ -1.9,\ 0.3,\ 0.1,\ 0.4,\ 0.1,\ 0.3,\ 0.1,\ -1.6,\ -1.6\}.$$

You then observed again the output of X and Y, obtaining two additional output sequences:

$$X_2 = \{0.5,\ 1.2,\ \ \ 3\},$$
$$Y_2 = \{1.1,\ 0.5,\ -1.4\}.$$

(a) Compute the quantized sequences $\tilde{X}_1 = Q(X_1)$ and $\tilde{Y}_1 = Q(Y_1)$ using the following scalar uniform quantizer:

$$a_1 = -\infty,\ a_2 = -1,\ a_3 = 1,\ a_4 = +\infty,$$

$$Q(x) = I_k,\ \text{if } x \in (a_k, a_{k+1}]. \qquad I_k = \begin{cases} a_{k+1} - \frac{\Delta}{2},\ \text{if } k = 1, \\ \frac{a_k + a_{k+1}}{2},\ \text{if } k = 2, \\ a_k + \frac{\Delta}{2},\ \text{if } k = 3. \end{cases}$$

(b) From $\tilde{X}_1$ and $\tilde{Y}_1$, estimate the joint p.m.f. $P(\tilde{X}; \tilde{Y})$ of a source that emits pairs of symbols $\{\tilde{X}(n), \tilde{Y}(n)\}$.

(c) Based on the estimated p.m.f., compute the entropies $H(\tilde{X})$, $H(\tilde{Y})$ and $H(\tilde{X}; \tilde{Y})$. From these quantities, derive the mutual information $I(\tilde{X}; \tilde{Y})$.

(d) Based on the estimated p.m.f., design the Huffman code that jointly encodes one symbol of $\tilde{X}$ and one symbol of $\tilde{Y}$. Compute the average Huffman code length and verify that it satisfies the theoretical bound.

(e) Using the same quantizer described in (a), compute the quantized sequences $\tilde{X}_2 = Q(X_2)$ and $\tilde{Y}_2 = Q(Y_2)$. Consider again a source that emits pairs of symbols $\{\tilde{X}(n), \tilde{Y}(n)\}$. How many bits are used to encode the information contained in $\tilde{X}_2$ and $\tilde{Y}_2$ using Huffman coding scheme.

Solution

(a) The defined quantizer is a scalar uniform one, thus $\Delta = \frac{a_3 - a_2}{2} = 2$. We can then compute the three reconstruction levels: $I_1 = a_2 - \frac{\Delta}{2} = -1 - 1 = -2$, $I_2 = \frac{a_2 + a_3}{2} = 0$ and $I_3 = a_3 + \frac{\Delta}{2} = 2$. At this point, we can apply the quantization rule $Q(x)$ obtaining

$$\tilde{X}_1 = \{-2,\ -2,\ -2,\ \ 0,\ 0,\ 0,\ \ 2,\ 2,\ 2,\ -2,\ -2,\ -2,\ 2,\ 2,\ 2,\ 0,\ 0,\ 0,\ \ 2,\ \ 0\},$$
$$\tilde{Y}_1 = \{-2,\ \ 0,\ \ 2,\ -2,\ 0,\ 2,\ -2,\ 0,\ 2,\ -2,\ -2,\ -2,\ 0,\ 0,\ 0,\ 0,\ 0,\ 0,\ -2,\ -2\}.$$

(b) To compute the joint pmf, we simply count the occurrences of the pairs $\{\tilde{X}(n), \tilde{Y}(n)\}$. As an example, the pair $\{-2, -2\}$ appears 4 times over 20 (i.e., for $n = 1, 10, 11, 12$), thus it has probability $4/20 = 0.2$. Following the same rule, we compute

$P(\tilde{X}; \tilde{Y})$	$\tilde{Y} = -2$	$\tilde{Y} = 0$	$\tilde{Y} = 2$
$\tilde{X} = -2$	0.2	0.05	0.05
$\tilde{X} = 0$	0.1	0.2	0.05
$\tilde{X} = 2$	0.1	0.2	0.05

By summing values in each row, we obtain the marginal probability of $\tilde{X}$: $P(\tilde{X} = -2) = 0.3$, $P(\tilde{X} = 0) = 0.35$, $P(\tilde{X} = 2) = 0.35$. By summing values in each column, we obtain the marginal probability of $\tilde{Y}$: $P(\tilde{Y} = -2) = 0.4$, $P(\tilde{Y} = 0) = 0.45$, $P(\tilde{Y} = 2) = 0.15$. Notice that $P(\tilde{X})$, $P(\tilde{Y})$ and $P(\tilde{X}; \tilde{Y})$ add to one.

(c) We apply the definitions

$$\begin{aligned} H(\tilde{X}) &= -\sum_{x=-2,0,2} P(\tilde{X} = x) \log_2(P(\tilde{X} = x)) = 1.5813, \\ H(\tilde{Y}) &= -\sum_{y=-2,0,2} P(\tilde{Y} = y) \log_2(P(\tilde{Y} = y)) = 1.4577, \\ H(\tilde{X}, \tilde{Y}) &= -\sum_{x=-2,0,2} \sum_{y=-2,0,2} P(\tilde{X} = x; \tilde{Y} = y) \log_2(P(\tilde{X} = x; \tilde{Y} = y)) = 2.9219. \end{aligned} \tag{1}$$

Notice that $H(\tilde{X}, \tilde{Y}) \leq H(\tilde{X}) + H(\tilde{Y})$ as it should be. Mutual information can be computed as $I(\tilde{X}, \tilde{Y}) = H(\tilde{X}) + H(\tilde{Y}) - H(\tilde{X}, \tilde{Y}) = 0.1171$. It is possible to notice that $\tilde{X}$ and $\tilde{Y}$ are not independent.

(d) We compute the Huffman code associated to each symbol $\{\tilde{X}, \tilde{Y}\}$. Results are reported in Table 1 and Figure 1. The average code length is $L_H = \sum_{x=-2,0,2} \sum_{y=-2,0,2} P(\tilde{X} = x; \tilde{Y} = y) l_h(\tilde{X} = x; \tilde{Y} = y) = 3$. The theoretical bound $H(\tilde{X}; \tilde{Y}) \leq L_H < H(\tilde{X}; \tilde{Y}) + 1$ is respected.

Symbol $\{\tilde{X}(n), \tilde{Y}(n)\}$	p.m.f.	Code	Length (l_H)
$\{-2, -2\}$	0.2	001	3
$\{-2, 0\}$	0.05	1001	4
$\{-2, 2\}$	0.05	1000	4
$\{0, -2\}$	0.1	111	3
$\{0, 0\}$	0.2	000	3
$\{0, 2\}$	0.05	1011	4
$\{2, -2\}$	0.1	110	3
$\{2, 0\}$	0.2	01	2
$\{2, 2\}$	0.05	1010	4

Table 1: Huffman code.

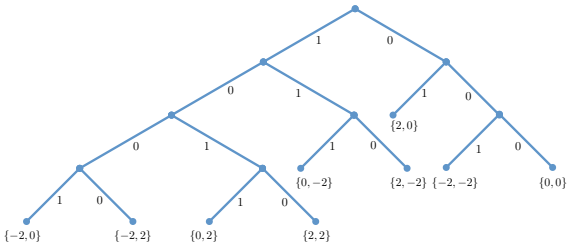


Figure 1: Huffman tree.

(e) Applying the given quantization rule we obtain $\tilde{X}_2 = \{0, 2, 2\}$ and $\tilde{Y}_2 = \{2, 0, -2\}$. The pairs $\{\tilde{X}(n), \tilde{Y}(n)\}$ to transmit are then $\{0, 2\}$, $\{2, 0\}$ and $\{2, -2\}$. The corresponding Huffman codes are 1011, 01 and 110. The total amount of bits is 9.

2. **Question (6 pts)**

(a) Explain which are the common visual irrelevancy and redundancy effects that can be exploited for image coding.

(b) Illustrate the block diagram of the JPEG compression coding scheme, detailing each block.

(c) Which are the advantages of using DCT with respect to DFT? And with respect to KLT?

Solution See lecture slides.

3. **Question (6 pts)**
   (a) Given a scalar uniform quantizer, derive the optimum reconstruction levels to minimize the error variance.
   (b) Illustrate the steps of the Lloyd-Max algorithm for scalar quantization, commenting the difference with respect to the optimum uniform quantizer described in (a).

   Solution See lecture slides.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**
   (a) Consider two discrete memoryless sources (DMS) X and Y
   - [T] [F] $H(X) < 1$ if X is binary.
     FALSE. It can be equal to 1.
   - [T] [F] $H(X) + H(Y) = H(X;Y)$ if X and Y are independent.
     TRUE.
   - [T] [F] $H(X) = 2$ means that each symbol of X can be represented with 2 bits.
     FALSE. It is an average value, thus some symbols may need more and some less.
   - [T] [F] $I(X;Y)$ cannot be computed for independent sources.
     FALSE. It can be computed and it is equal to 0.

   (b) Consider a transform coding scheme $\mathbf{y} = \mathbf{T}\mathbf{x}$, where $\mathbf{T}$ is the matrix representing a unitary transform, $\mathbf{x}$ is the data vector, and $\mathbf{y}$ contains the transform coefficients
   - [T] [F] Transform coding gain is always positive.
     FALSE. It can be equal to 0.
   - [T] [F] To increase coding gain it is better to use a transform that grants uniformly distributed variance among the transform coefficients.
     FALSE. It is the opposite. This situation gives 0 gain.
   - [T] [F] The optimal amount of bits for some transform coefficient may be theoretically negative.
     TRUE

   (c) Consider an audio coding system that exploits perceptual coding
   - [T] [F] A tone at time t_2 can mask signal components at time $t_1 < t_2$.
     TRUE.
   - [T] [F] Critical bandwidths are linearly distributed in the Bark scale.
     TRUE.
   - [T] [F] Perceptual coding is used for lossless audio compression.
     FALSE. If some component of the signal (e.g., those not perceived by human hearing system) are removed by coding, it is not possible to perfectly (i.e., lossless) reconstruct the original signal from the encoded one.

5. **Question (Matlab - 10pts)**
   (a) Load the file `data.mat`, which contains the monodimensional signal `y` and the dictionary `dict` composed by 8 words (i.e., 8 rows) of 4 samples each.
   (b) Encode `y` using PCM with a uniform scalar quantizer with quantization intervals $\Delta \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$.

(c) Encode `y` using a closed-loop DPCM scheme that works on blocks of 4 samples. The predictor of each block is the previous reconstructed block. The first block is transmitted using PCM. Use quantization intervals $\Delta \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$.

(d) Encode `y` using a custom predictive coding scheme that works on blocks of 4 samples. Each block is predicted using the closest word in the dictionary in the MSE sense. The decoder knows the dictionary. For each block, the encoder sends to the decoder: (i) the index $w \in \{1, ..., 8\}$ of the selected word; (ii) the quantized prediction residuals. Quantize prediction residuals using $\Delta \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$ as quantization intervals.

(e) For each case (i.e., (b), (c), (d)) plot the rate-distortion (RD) curves, using MSE as noise metric, SNR in decibels as distortion, and assuming that you are able to represent quantized coefficients using the minimum amount of bits. For point (c) assume that you are able to transmit the word indexes w with a lossless scheme that uses the minimum amount of bits.

(f) Comment the obtained RD curves (hint: keep in mind the DPCM coding gain equation).

Solution

```
clc, clear all, close all

%% Load data
load('data.mat', 'dict', 'y')

%% Parameters and initialization
delta_all = [.1 .2 .3 .4 .5];
block = size(dict,2);
y_len = length(y);

MSE_pcm = zeros(length(delta_all),1);
H_pcm = zeros(length(delta_all),1);

MSE_dpcm = zeros(length(delta_all),1);
H_dpcm = zeros(length(delta_all),1);

MSE_dict = zeros(length(delta_all),1);
H_dict = zeros(length(delta_all),1);

%% PCM
for d = 1:length(delta_all)

    %%% Quantization step
    delta = delta_all(d);

    %%% PCM
    y_tilde = delta * round(y/delta);
    MSE_pcm(d) = sum( (y-y_tilde).^2 );

    %%% Entropy
    h = hist(y_tilde, min(y_tilde):delta:max(y_tilde));
    h = h(h~=0);
    p = h./sum(h);
    H_pcm(d) = -sum(p.*log2(p));

end
SNR_pcm = pow2db(sum(y.^2)./MSE_pcm);

%% DPCM
for d = 1:length(delta_all)

    %%% Initialize variables
    y_tilde = nan*zeros(size(y));
    err_q = nan*zeros(size(y));

    %%% Quantization step
    delta = delta_all(d);

    %%% PCM
    y_tilde(1:block) = delta*floor(y(1:block)/delta);

    %%% DPCM
    for i = block+1:block:length(y_tilde)-block+1
        err = y(i:i+block-1) - y_tilde(i-block:i-1);
        err_q(i:i+block-1) = delta * round(err/delta);
        y_tilde(i:i+block-1) = y_tilde(i-block:i-1) + err_q(i:i+block-1);
    end
    MSE_dpcm(d) = sum( (y-y_tilde).^2 );
```

```
    %%% Entropy
    err_q(1:block) = y_tilde(1:block);
    h = hist(err_q, min(err_q):delta:max(err_q));
    h = h(h~=0);
    p = h./sum(h);
    H_dpcm(d) = -sum(p.*log2(p));

end
SNR_dpcm = pow2db(sum(y.^2)./MSE_dpcm);


%% DPCM-dict
for d = 1:length(delta_all)
    delta = delta_all(d);

    %%% Init
    y_tilde = nan*zeros(size(y));
    err_q = nan*zeros(size(y));
    pred_idx = nan*zeros(length(y)/block,1);

    %%% DPCM
    for i = 1:block:length(y_tilde)-1
        %%% find predictor
        err_pred = dict' - repmat(y(i:i+block-1),1,size(dict,1));
        MSE_pred = sum(err_pred.^2);
        [~, pred_idx((i+block-1)/block)] = min(MSE_pred);

        %%% apply prediction
        pred = dict(pred_idx((i+block-1)/block),:)';
        err = y(i:i+block-1) - pred;
        err_q(i:i+block-1) = delta * round(err/delta);
        y_tilde(i:i+block-1) = pred + err_q(i:i+block-1);
    end
    MSE_dict(d) = sum( (y-y_tilde).^2 );

    %%% Entropy err
    h = hist(err_q, min(err_q):delta:max(err_q));
    h = h(h~=0);
    p = h./sum(h);
    H_dict(d) = -sum(p.*log2(p));

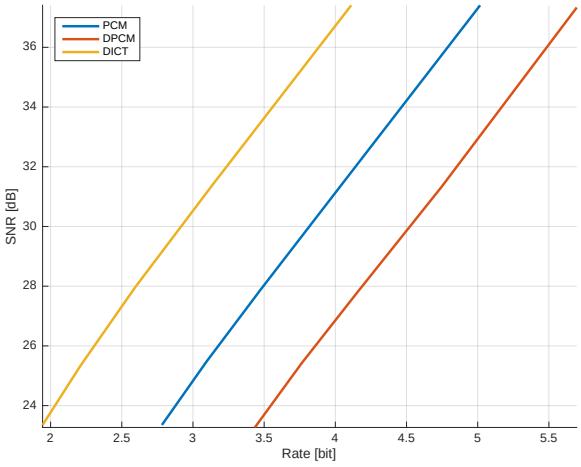
    %%% Entropy dict
    h = hist(pred_idx, min(pred_idx):1:max(pred_idx));
    h = h(h~=0);
    p = h./sum(h);
    H_dict(d) = H_dict(d)-sum(p.*log2(p))/block; % we transmit 1 dictionary index every 4 symbols

end
SNR_dict = pow2db(sum(y.^2)./MSE_dict);


%% Plot RD curves
figure(2), clf, hold on
plot(H_pcm, SNR_pcm, 'linewidth', 2)
plot(H_dpcm, SNR_dpcm, 'linewidth', 2)
plot(H_dict, SNR_dict, 'linewidth', 2)
grid on, axis tight, xlabel('Rate [bit]'), ylabel('SNR [dB]')
legend('PCM', 'DPCM', 'DICT', 'Location', 'NorthWest')

% Comment 1: DPCM distrotion is greater than PCM one as the use of the
% previous block as prediction is a bad choice. As a matter of fact, it is
% easy to verify that the variance of the residual (err) is greater than
% the variance of the signal (y).

% Comment 2: Using the custom prediction method based on a dictionary, the
% variance of the residual (err) is very low, thus enabling to achieve low
% quantization error. As a matter of fact, even considering the additional
% transmitted information (i.e., a word index every four samples), this
% method outperforms both PCM and DPCM. Clearly this strongly depends on
% the used dictionary and signal.
```


PCM
DPCM
DICT
SNR [dB]
Rate [bit]
24
26
28
30
32
34
36
2
2.5
3
3.5
4
4.5
5
5.5